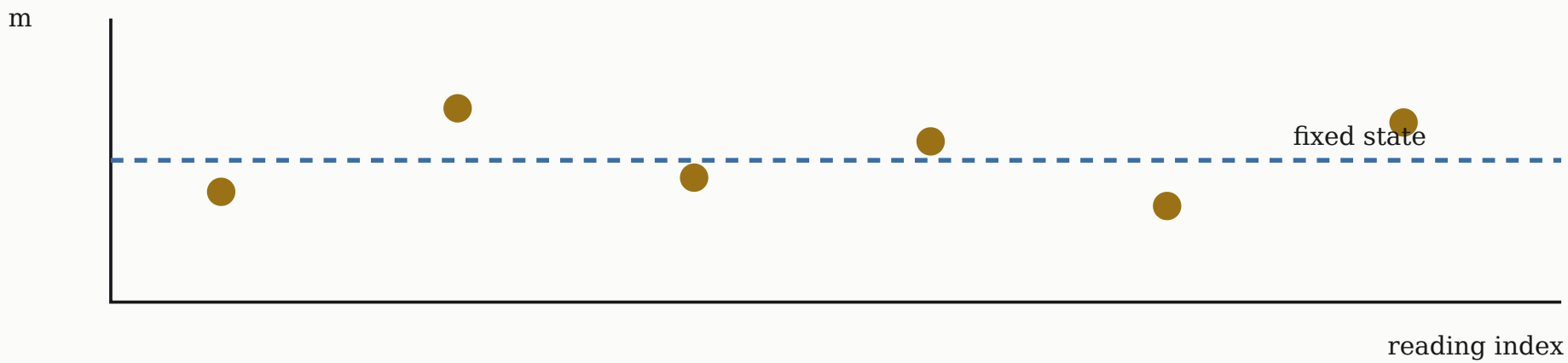
From Noisy Readings to Reliable Estimates

Which position should a noisy sensor report?



> One stationary object. Several different readings.

`state` · `measurements` · `inferred answer`

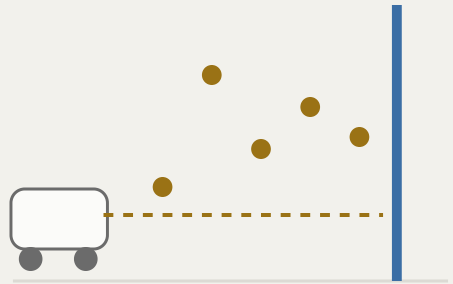
Report an estimate and the uncertainty around it.

LECTURE ROADMAP

From Readings to Estimates

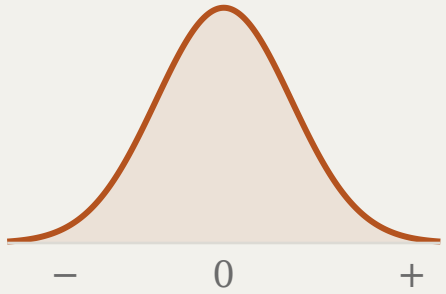
How does the lecture fit together?

Readings



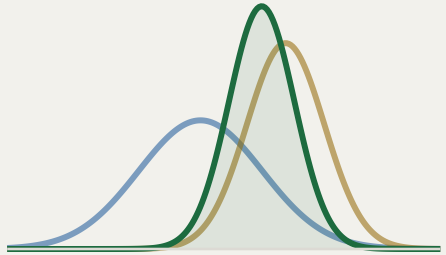
Same wall, noisy readouts

Error



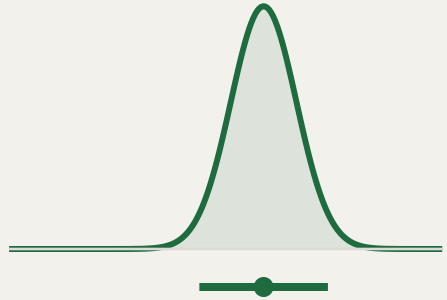
Model the error

Update



Combine information

Estimate



Report a value and a range

> What should we report? How sure are we?

Probability distribution; uncertainty — [1] DL §3.1 · Course concept map

Random variables

How does an outcome become a number?

$$x : \Omega \to \mathbb{R}, \qquad x(\omega) \in \{-1, 0, 2\}$$

- Ω sample space: all possible outcomes
- ω one outcome
- x random error (cm)

Outcome A $\to -1$ cm

Outcome B $\to 0$ cm

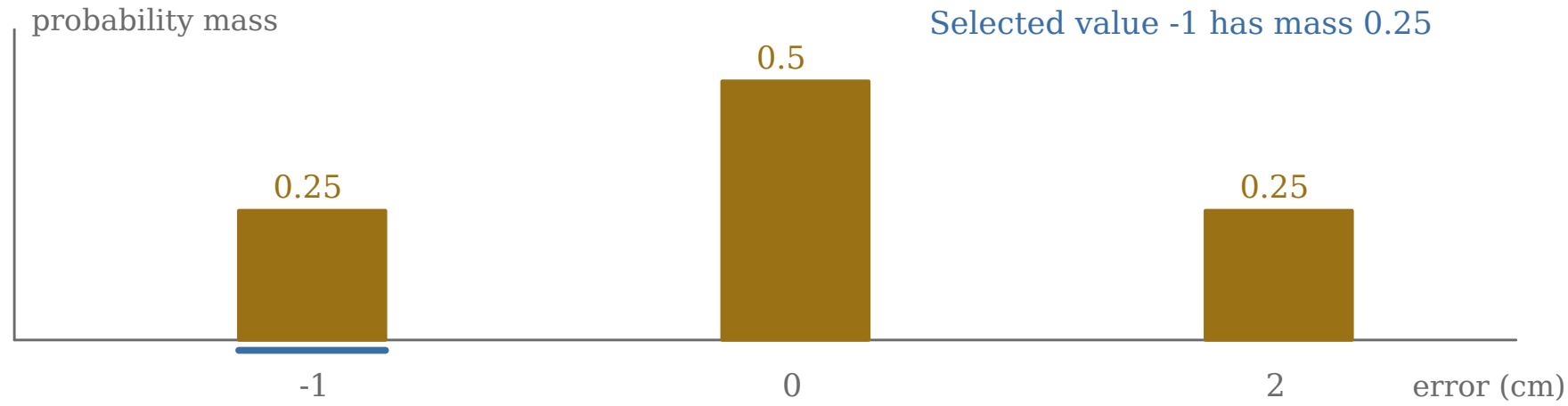
Outcome C $\to 2$ cm

> The rule is fixed; the outcome is uncertain.

Random variable; realization — [1] DL §3.2

Probability Mass Function (PMF)

How do probabilities attach to discrete values?



$$p(a) = P(x = a), \qquad p(a) \geq 0, \qquad \sum_{a} p(a) = 1$$

x random error (cm) a possible value (cm)

$p(a)$ probability of $x = a$ P event probability

Probability mass function (PMF) — [1] DL §3.3.1

DEFINITION AND REASONING

Probability Mass Function (PMF)

What does the three-point error model predict?

Error values -1, 0, 2 cm; masses ¼, ½, ¼. Find nonnegative-error probability and mean.

$$P(x \geq 0) = \tfrac{1}{2} + \tfrac{1}{4} = \tfrac{3}{4}, \qquad \mu = (-1)\tfrac{1}{4} + 0\tfrac{1}{2} + 2\tfrac{1}{4} = \tfrac{1}{4}$$

- P event probability
- x error (cm)
- μ mean error (cm)

Discrete probability; expectation — [1] DL §3.3.1 · [1] DL §3.8 · Course example

CHOOSE, EXPLAIN, VOTE

Probability Mass Function (PMF)

Which values belong to the event?

Errors $-1, 0, 2$ cm have masses $1/2, 1/4, 1/4$. What is $P(x \geq 0)$?

A $1/4$: count only positive errors.

B $1/2$: include zero and positive errors.

C $2/3$: two of three possible values.

Event probability; PMF — [1] DL §3.3.1 · Course question

Probability Density Function (PDF)

Why does a continuous event need an area?

$$p(t) \geq 0, \quad \int_{-\infty}^{\infty} p(t)\,dt = 1$$

$$P(a \leq x \leq b) = \int_{a}^{b} p(t)\,dt$$

- x random error (cm)
- a, b interval endpoints
- t integration variable
- $p(t)$ density (cm^{-1})

$$P(x = c) = \int_{c}^{c} p(t)\,dt = 0$$

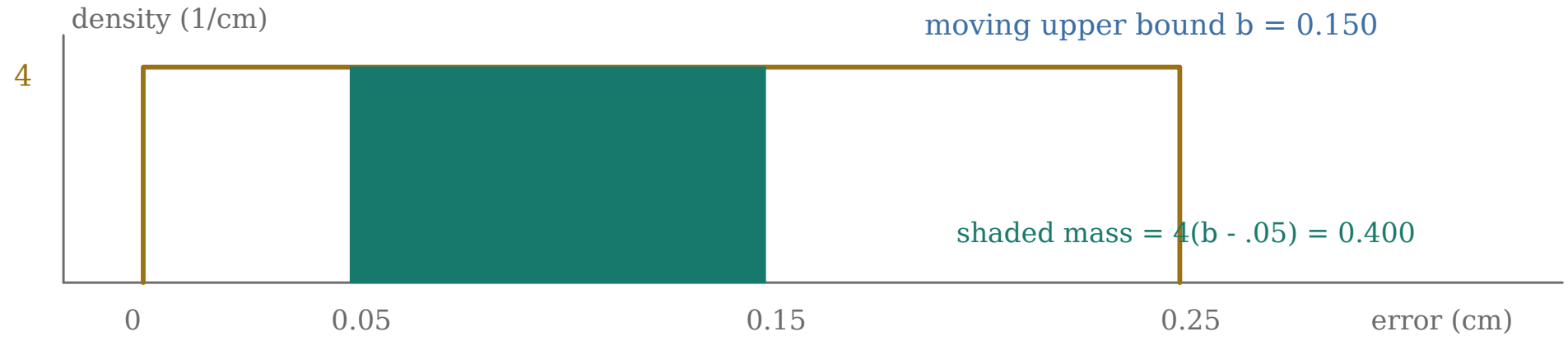
- c fixed error value

Probability density function (PDF) — [1] DL §3.3.2

DEFINITION AND REASONING

Probability Density Function (PDF)

Can a valid density be greater than one?



UNIFORM ERROR

$$p(t) = 4 \ \ (0 \le t \le .25), \quad P(.05 \le x \le .15) = 4(.15 - .05) = .4$$

- t, x error (cm)
- p density (cm^{-1})
- P interval probability

Uniform density; normalization — [1] DL §3.3.2 · Course example

CHOOSE, EXPLAIN, VOTE

Probability Density Function (PDF)

Density is height; probability is area

x is uniform on $[0, .5]$ cm. Which statement is valid?

A. Density 2 cm^{-1}; $P(.1 \leq x \leq .3) = .4$.

B. Density 2 means event probability 2.

C. Probability is $.2$, the interval width alone.

Density; event probability — [1] DL §3.3.2 · Course question

Moments

How do we average a function of a random value?

$$\mathbb{E}[g(x)] = \sum_{a} g(a)p(a)$$

- x discrete random variable
- a possible value
- g function being averaged
- p PMF

$$\mathbb{E}[g(x)] = \int_{-\infty}^{\infty} g(t)p(t)\,dt$$

- x continuous random variable
- t integration variable
- p density
- g assume E[|g(x)|] is finite

Expectation of a function — [1] DL §3.8 · Course derivation

CHOOSE, EXPLAIN, VOTE

Moments

What does centering remove?

$\mathrm{Var}(x) = \mathbb{E}[(x-\mu)^2]$. Given $\mu = 1/4$ cm and $\mathbb{E}[x^2] = 5/4$ cm^2, choose the variance.

A $5/4$: the second moment is always variance.

B 1: subtract the mean, not its square.

C $19/16$: subtract the squared mean.

Centering; variance — [1] DL §3.8 · Course question

Moments

What is the mean of a random column vector?

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \qquad \mu = \mathbb{E}[x] = \begin{bmatrix} \mathbb{E}[x_1] \\ \mathbb{E}[x_2] \end{bmatrix}$$

- x random column in $\mathbb{R}^2$
- x_i component i
- μ mean column in $\mathbb{R}^2$

> Average corresponding components; preserve the column shape.

Samples use superscripts: $x^{(s)}$. Components use subscripts: x_i.

Random vector; mean vector — [1] DL §3.8

DEFINITION AND REASONING

Moments

Why does covariance use an outer product?

$$\Sigma = \mathbb{E}[(x-\mu)(x-\mu)^T], \qquad \Sigma_{ij} = \mathbb{E}[(x_i-\mu_i)(x_j-\mu_j)]$$

x, μ random column and mean in R^{d}

Σ population covariance, d × d

i, j coordinate indices

> Diagonal: component variance. Off-diagonal: paired deviations.

Covariance matrix — [1] DL §3.8 · Course derivation

Moments

Why is covariance positive semidefinite?

$$a^T \Sigma a = \mathbb{E}[(a^T(x-\mu))^2] \geq 0$$

- a fixed direction column in R^{d}
- x, μ random column and population mean
- Σ population covariance: PSD means $a^T \Sigma a \geq 0$ for every a

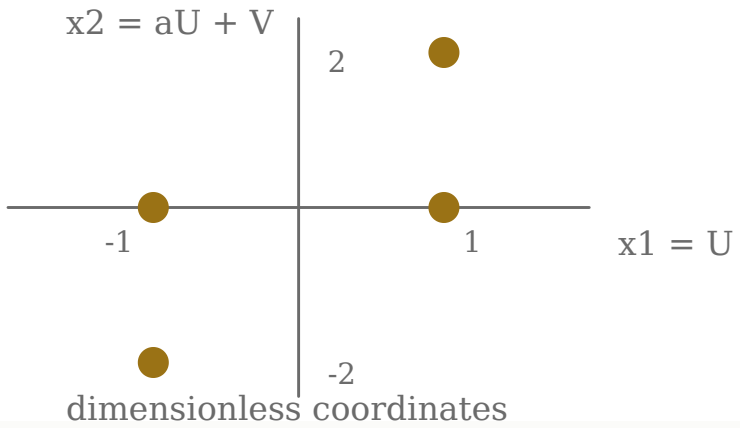
$$C = \frac{X_c^T X_c}{N-1} \quad (N > 1)$$

- C sample covariance estimate
- X_c N centered sample rows
- N number of samples

Positive semidefinite covariance — [1] DL §3.8 · Course derivation

Moments

How do independent signs create correlation?



a = 1.00

four equally likely outcomes

a: 0 to 1 adds the shared sign

Independent fair signs U,V: knowing one does not change the other's probabilities. Let $x = (U, U+V)^T$.

$$\mu = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \qquad \Sigma = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$$

U, V independent, equally likely ± 1

x, μ random column and mean

Σ population covariance

Independence; covariance — [1] DL §3.7 · [1] DL §3.8 · Course example

CHOOSE, EXPLAIN, VOTE

Moments

Independent sources, dependent components

U, V are independent fair signs. Let $x = (U, 2U + V)^T$. What is $\mathrm{Cov}(x_1, x_2)$?

A. 2: both components share U.

B. 0: independent sources imply independent outputs.

C. 1: doubling a coefficient has no effect.

Independence; covariance — [1] DL §3.7 · [1] DL §3.8 · Course question

DEFINITION AND REASONING

Moments

Does zero covariance guarantee independence?

x is uniform on $\{-1,0,1\}$; $y=x^2$. Is y independent of x?

$$\operatorname{Cov}(x,y)=0, \qquad P(y=0 \mid x=0)=1 \neq \tfrac{1}{3}=P(y=0)$$

x, y	random scalars
P	event probability
Cov	population covariance

Uncorrelatedness; independence — [1] DL §3.8 · [2] UBC Lec.18 p.8 · Course example

Gaussian geometry

What defines a multivariate Gaussian density?

$$p(x) = (2\pi)^{-d/2} |\Sigma|^{-1/2} \exp(-d_M^2/2)$$

- x, μ evaluation point and mean in R^d
- d dimension
- $\Sigma \succ 0$ positive variance in every nonzero direction
- $|\Sigma|$ determinant
- p density
- d_M^2 squared Mahalanobis distance

$$d_M^2 = (x-\mu)^T \Sigma^{-1} (x-\mu)$$

- Σ^{-1} inverse covariance
- $x - \mu$ centered displacement

Multivariate Gaussian distribution — [1] DL §3.9.3

Gaussian geometry

Change basis, then standardize

Derivation

Derive the mean and covariance

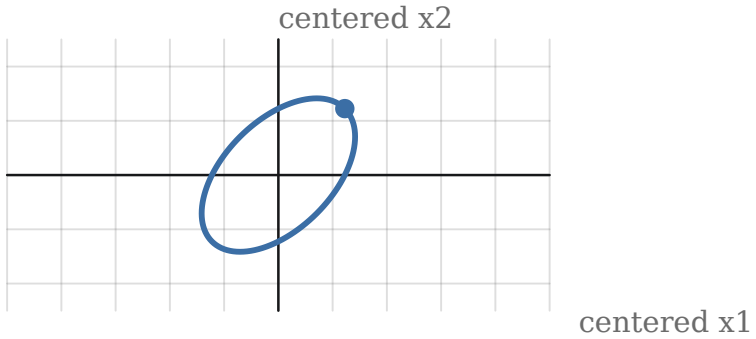
$\mu = \mathbb{E}[x]$; $\Sigma = \mathrm{Cov}(x) = Q\Lambda Q^T \succ 0$; $Q^TQ = I$; Λ positive diagonal. Fixed whitening map $W = \Lambda^{-1/2}Q^T$; $z = W(x-\mu)$; I: identity.

1. $\mathbb{E}[z] = W(\mathbb{E}[x] - \mu) = 0$ by linearity.
2. $\mathrm{Cov}(z) = \mathbb{E}[zz^T] = W\,\mathbb{E}[(x-\mu)(x-\mu)^T]\,W^T = W\Sigma W^T$.
3. $= \Lambda^{-1/2}Q^T(Q\Lambda Q^T)Q\Lambda^{-1/2} = \Lambda^{-1/2}\Lambda\Lambda^{-1/2} = I$.
4. Fixed matrices pass through expectation; $Q^TQ = I$ cancels the basis changes.

Whitening; eigendecomposition — [2] UBC Lec.18 p.5 · Course derivation

Gaussian geometry

A contour radius becomes an ellipse axis



unit Mahalanobis contour

eigenvalues 3 and 1

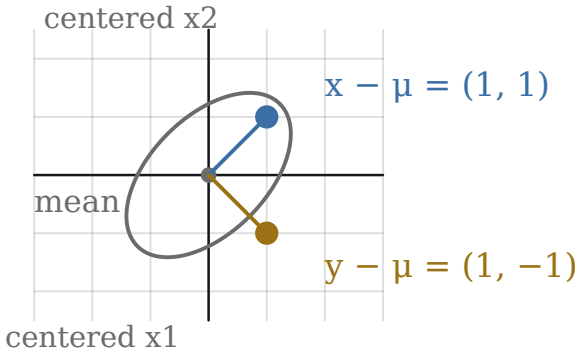
$$x = \mu + Q\Lambda^{1/2}z, \qquad \|z\| = c$$

c standardized radius; q_i: unit eigenvectors (columns of Q); $\sqrt{\lambda_i}$: axis scale. Plot of $x - \mu$: $c = 1$, $\lambda = (3, 1)$, $q_1 = (1, 1)^T/\sqrt{2}$, $q_2 = (1, -1)^T/\sqrt{2}$.

Gaussian density contour — [2] UBC Lec.18 p.5 · Course derivation

Gaussian geometry

Equal lengths can have unequal surprise



gray ellipse: $d_m = 1$

both arrows: Euclidean length $\sqrt{2}$

$$\Sigma = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad \mu = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

Σ covariance; μ mean; compare $x = (2, 0)^T$ and $y = (2, -2)^T$.

$$d_M^2(x) = \tfrac{2}{3}, \qquad d_M^2(y) = 2$$

d_M covariance-scaled distance.

Mahalanobis distance — [2] UBC Lec.18 p.5 · Course example

CHOOSE, EXPLAIN, VOTE

Gaussian geometry

Which direction makes a point less surprising?

$\mu = 0$, $\Sigma = \mathrm{diag}(4, 1)$; $u = (2, 0)^T$, $v = (0, 2)^T$. Compare Gaussian densities.

A $p(u) = p(v)$: their Euclidean lengths agree.

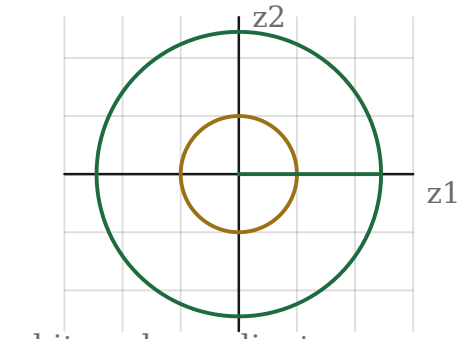
B $p(u) > p(v)$: u is one standard deviation away.

C $p(u) < p(v)$: higher variance always lowers density.

Mahalanobis distance — [2] UBC Lec.18 p.5 · Course question

Gaussian geometry

A unit ellipse contains less than half



c = 1: mass 39.3%

c = 2.448: mass 95%

original semiaxes: 4.24 and 2.45

circle radius = Mahalanobis distance

$$P(d_M^2 \le c^2) = 1 - e^{-c^2/2} \quad (d = 2)$$

c contour radius; $d = 2$ dimension; the formula assumes a nonsingular Gaussian. Example: $\lambda_1 = 3, \lambda_2 = 1$.

$$c_{.95} = \sqrt{-2 \log .05} \approx 2.44775$$

$c_{.95}$ radius enclosing 95%.

Chi-square distribution; Gaussian probability region — [3] NIST §1.3.6.6.6 · [3] NIST χ^2 table · Course derivation

Bayes

Which population does a positive select?

joint events	positive +	negative −
event H	H and +	H and −
no event	not H and +	

$$P(H \mid +) = \frac{P(H \cap +)}{P(+)}$$

$$P(+ \mid H) = \frac{P(H \cap +)}{P(H)}$$

H: event; $+$: positive reading; $\cap$: both occur. Each denominator must be positive.

Conditional probability — [1] DL §3.4–3.5 · Course example

Bayes

Write the joint event in two ways

$$P(H \mid +) = \frac{P(+ \mid H)P(H)}{P(+)}$$

$P(H)$ prior; $P(+ \mid H)$ likelihood; $P(+) > 0$ evidence. H: event; $+$: positive reading.

Bayes' rule — [1] DL §3.11 · Course derivation

Bayes

Normalize over both sources of positives

$$P(+) = P(+ \mid H)P(H) + P(+ \mid \neg H)P(\neg H)$$

$\neg H$ complement; the two joint events are disjoint and exhaustive. H: event; $+$: positive reading.

Law of total probability; evidence — [1] DL §3.4–3.5 · [1] DL §3.11 · Course derivation

Bayes

Why can most positive alarms be false?

joint events	positive +	negative −
event H	90	10
no event	990	8910

$$P(H) = .01, \quad P(+ \mid H) = .9, \quad P(+ \mid \neg H) = .1$$

10,000 model trials: 100 event cases and 9,900 non-event cases.

$$P(H \mid +) = \frac{90}{90 + 990} = \frac{1}{12}$$

H event; $+$ positive reading.

Prior probability; posterior probability — [1] DL §3.11 · Course example

CHOOSE, EXPLAIN, VOTE

Bayes

How does prevalence change the chance an alarm is true?

$P(H) = .1$, $P(+ \mid H) = .9$, $P(+ \mid \neg H) = .1$. What is $P(H \mid +)$?

A .9: the chance of a true event equals sensor sensitivity.

B .1: an observation cannot change the prior.

C .5: true and false positives contribute equally.

Prior probability; posterior probability — [1] DL §3.11 · Course question

Break

Pause: probability, density, and conditioning

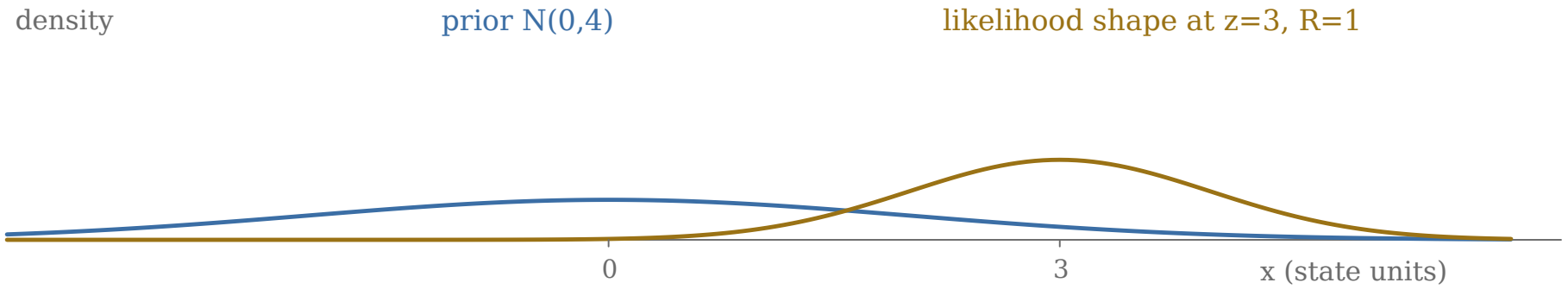
$$P(X \in A), \qquad p(x), \qquad P(H \mid +)$$

X random quantity; x evaluation value; A event; p density. H: event of interest; $+$: positive reading. Conditioning changes the reference population.

Event probability; density; conditional probability — [1] DL §3.3.2 · [1] DL §3.4–3.5 · Course recap

Gaussian conditioning

A numeric reading updates a curve



$$x \sim \mathcal{N}(\mu_0, P_0), \quad z = x + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, R)$$

x: state; z: reading; $P_0, R > 0$: prior/noise variances; ε independent of x. $p(x)$: prior; $p(z \mid x)$: likelihood at fixed z; $p(x \mid z)$: posterior.

Gaussian prior; Gaussian likelihood — [2] UBC Lec.18 p.7

Gaussian conditioning

Add precisions, then complete the square

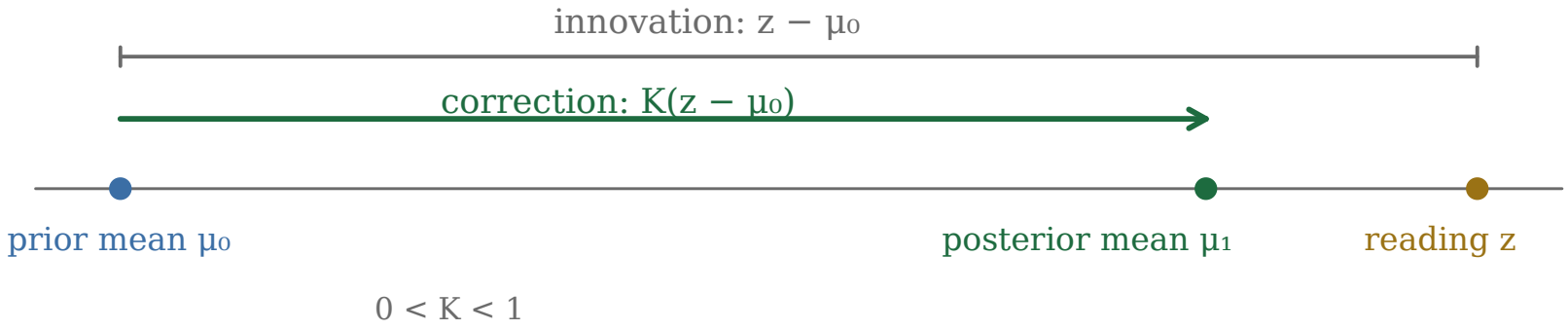
$$J(x) = \frac{(x-\mu_0)^2}{2P_0} + \frac{(z-x)^2}{2R}, \quad P_1^{-1} = P_0^{-1} + R^{-1}, \quad \mu_1 = P_1(\mu_0/P_0 + z/R)$$

J: negative log posterior, up to a constant. Independent Gaussian prior/noise; $P_0, R > 0$. x: state; z: reading. μ_0, P_0: prior mean/variance; R: noise variance; μ_1, P_1: posterior mean/variance.

Gaussian posterior; precision — [2] UBC Lec.18 p.7 · [1] DL §3.9.3 · Course derivation

Gaussian conditioning

The gain weights the innovation



$$\mu_1 = \frac{R\mu_0 + P_0 z}{P_0 + R} = \mu_0 + K(z - \mu_0), \quad K = \frac{P_0}{P_0 + R}, \quad P_1 = (1 - K)P_0$$

μ_0, P_0: prior mean/variance; z: reading; R: noise variance. μ_1, P_1: posterior mean/variance. Innovation $z - \mu_0$; gain $0 < K < 1$.

Kalman gain; measurement innovation — [5] Welch–Bishop §1 · [2] UBC Lec.18 p.7 · Course derivation

Gaussian conditioning

Fuse a prior and a precise reading



$$\mu_0 = 0, \quad P_0 = 4, \quad z = 3, \quad R = 1$$

μ_0: prior mean; P_0: prior variance; z: reading; R: independent noise variance.

$$K = \tfrac{4}{5}, \qquad \mu_1 = 2.4, \qquad P_1 = .8$$

K gain; μ_1, P_1 posterior mean, variance.

Gaussian conditioning — [2] UBC Lec.18 p.7 · Course example

CHOOSE, EXPLAIN, VOTE

Gaussian conditioning

What does a noisier reading change?

Prior $\mathcal{N}(0, 4)$; $z = 3$ with independent Gaussian noise, variance R: $1 \to 4$. Compare posterior mean and variance.

A. Mean moves toward 0; posterior variance increases.

B. Mean moves toward 3; posterior variance decreases.

C. Only the mean changes; variance ignores sensor noise.

Gaussian conditioning — [2] UBC Lec.18 p.7 · Course question

Gaussian conditioning

A correlated block predicts another block

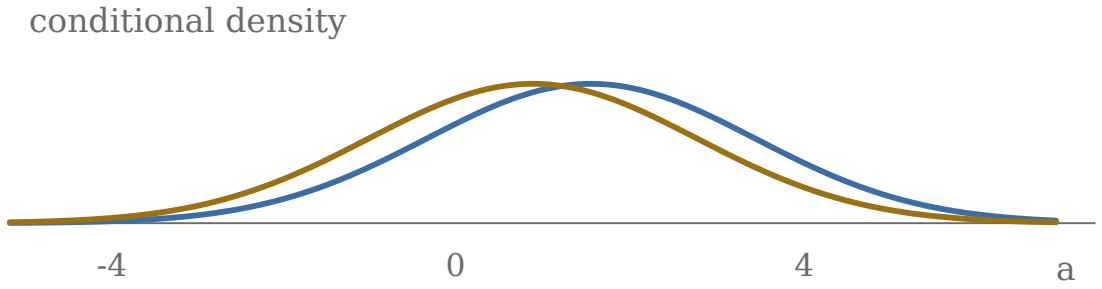
$$\mu_{a|b} = \mu_a + \Sigma_{ab}\Sigma_{bb}^{-1}(b - \mu_b)$$

Jointly Gaussian $a \in \mathbb{R}^p$, $b \in \mathbb{R}^q$; observe b. μ_a, μ_b: means; Σ_{bb}: invertible covariance of b. Cross-covariance: $\Sigma_{ab} = \mathbb{E}[(a - \mu_a)(b - \mu_b)^T]$.

Conditional Gaussian mean — [2] UBC Lec.18 p.7

Gaussian conditioning

Conditioning removes predictable variance



b = 5: mean 5/3

b = 2: mean 1

both variances 32/9

$$\Sigma_{a|b} = \Sigma_{aa} - \Sigma_{ab}\Sigma_{bb}^{-1}\Sigma_{ba}$$

a: unobserved block; b: observed block. Σ_{aa}: covariance of a; $\Sigma_{a|b}$: covariance after observing b. Joint Gaussian, $\Sigma_{bb} \succ 0$, $\Sigma_{ba} = \Sigma_{ab}^T$. Plot: $\mu = (1, 2)^T$, $\Sigma = \begin{bmatrix} 4 & 2 \\ 2 & 9 \end{bmatrix}$.

Conditional Gaussian covariance — [2] UBC Lec.18 p.7

CHOOSE, EXPLAIN, VOTE

Gaussian conditioning

Does the conditional variance change?

Gaussian $(a, b)^T$: $\mu = (1, 2)^T$, $\Sigma = \begin{bmatrix} 4 & 2 \\ 2 & 9 \end{bmatrix}$. Observe $b = -1$ instead of 5.

A Both conditional mean and variance decrease.

B Mean decreases; conditional variance stays fixed.

C Neither changes because the joint model is fixed.

Conditional Gaussian distribution — [2] UBC Lec.18 p.7 · Course question

Estimation

Maximum Likelihood Estimation (MLE)

$$b = A\theta + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2 I)$$

$$L(\theta; b) = p(b \mid \theta), \quad \hat{\theta}_{MLE} \in \operatorname*{argmax}_{\theta} L(\theta; b)$$

$b \in \mathbb{R}^N$: fixed data; $A \in \mathbb{R}^{N \times p}$: known; $\theta \in \mathbb{R}^p$: unknown; $\sigma^2 > 0$: known variance. L: likelihood; argmax: maximizer(s).

Gaussian linear regression; maximum likelihood — [1] DL §5.5.1

Estimation

Gaussian products become a squared loss

$$-\log p(b \mid \theta) = \frac{N}{2}\log(2\pi\sigma^2) + \frac{\|b - A\theta\|^2}{2\sigma^2}$$

A known; b observed; θ unknown; independent errors, each $\mathcal{N}(0, \sigma^2)$, $\sigma^2 > 0$.

Negative log-likelihood; least squares — [1] DL §5.5.1 · Course derivation

Estimation

The familiar normal equation returns

$$A^T A\hat{\theta} = A^T b, \qquad A^T(b - A\hat{\theta}) = 0$$

$A \in \mathbb{R}^{N\times p}$: known design; $b \in \mathbb{R}^N$: data; $\theta \in \mathbb{R}^p$: parameters. Independent equal-variance Gaussian errors make $\hat{\theta}$ the MLE. Full column rank gives uniqueness; residual is orthogonal to columns.

Normal equations; least squares — [1] DL §5.1.4 · Course derivation

Estimation

Fit three readings with two coefficients

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

$\theta = (\text{intercept}, \text{slope})^T$; $r = b - A\hat{\theta}$.

$$\hat{\theta} = \begin{pmatrix} 7/6 \\ 1/2 \end{pmatrix}, \quad r = \begin{bmatrix} -1/6 \\ 1/3 \\ -1/6 \end{bmatrix}$$

$\hat{\theta}$ estimate; r observed minus fitted.

Least-squares estimate — [1] DL §5.1.4 · Course example

Estimation

Maximum a Posteriori Estimation (MAP)

$$J(\theta) = \tfrac{1}{2}\|b - A\theta\|^2 + \tfrac{\lambda^2}{2}\|\theta - \mu_0\|^2, \quad \lambda^2 = \sigma^2/\tau^2$$

$\theta \sim \mathcal{N}(\mu_0, \tau^2 I)$ isotropic prior: variance τ^2 in every unit direction; σ^2 noise variance; λ^2 relative strength.

Maximum a posteriori (MAP); quadratic regularization — [1] DL §5.6.1 · Course derivation

Estimation

The prior changes the normal equation

$$(A^TA + \lambda^2 I)\hat{\theta} = A^Tb + \lambda^2\mu_0$$

Example: $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}$, $b = (1, 2, 2)^T$, $\mu_0 = 0$, $\lambda^2 = 1$.

$$\hat{\theta}_{MAP} = \begin{pmatrix} 4/5 \\ 3/5 \end{pmatrix}, \quad \|b - A\hat{\theta}\|^2 = \tfrac{2}{5}$$

$\hat{\theta}_{MAP} = (\text{intercept}, \text{slope})^T$: posterior mode; $\|b - A\hat{\theta}\|^2$: squared residual norm.

MAP normal equations; ridge regularization — [1] DL §5.6.1 · [1] DL §5.1.4 · Course derivation

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Estimation

What makes the MAP estimate unique?

$A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$, $b = (1, 2)^T$; $J = \frac{1}{2}\|b - A\theta\|^2 + \frac{1}{2}\|\theta\|^2$.

A. The prior makes the two data columns independent.

B. Gaussian noise makes every data fit unique.

C. The prior adds curvature along the data null space.

MAP estimate; positive definiteness — [1] DL §5.6.1 · Course question

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Integrated board

Calibration: choose the likelihood

$b = (1, 2, 3)^T$, $b_i = \theta + \varepsilon_i$, $\varepsilon_i \overset{iid}{\sim} \mathcal{N}(0, 1)$. Which data objective fits this model?

- **A** $\frac{1}{2}\sum_i (b_i - \theta)^2$
- **B** $\sum_i |b_i - \theta|$
- **C** $\frac{1}{2}\sum_i (b_i - \theta)^2 + \frac{1}{2}\theta^2$

Gaussian likelihood; maximum likelihood — [1] DL §5.5.1 · Course example

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Integrated board

Calibration: combine data and prior

Readings $(1, 2, 3)$; independent normal noise, variance 1; prior $\theta \sim \mathcal{N}(0, 1)$. Find posterior mean and precision.

A. Center stays at 2; prior changes only uncertainty.

B. Center moves toward 0; precision becomes 4.

C. Center moves toward 0; variances add to 4.

Gaussian posterior; MAP estimate — [2] UBC Lec.18 p.7 · [1] DL §5.6.1 · Course example

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Intervals

Calibration: confidence coverage

Three iid $B_i \mid \theta \sim \mathcal{N}(\theta, 1)$; $\bar{B}$: mean. $C(B) = \bar{B} \pm 1.96/\sqrt{3}$. Interpret 95% confidence.

A. This realized interval has 95% posterior mass.

B. 95% of future individual readings lie inside it.

C. Repeated intervals cover fixed θ 95% of the time.

Confidence interval; coverage probability — [3] NIST §7.1.4 · Course example

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Intervals

Calibration: what does credibility describe?

Observed $b = (1, 2, 3)^T$; $\theta \mid b \sim \mathcal{N}(1.5, .25)$. A 95% credible interval describes which probability?

A 95% posterior mass for θ, with b fixed.

B 95% coverage for every fixed true offset.

C 95% probability for a future noisy reading.

Posterior credible interval — [4] MIT 18.05 §2 · Course example

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Intervals

Calibration: why do the intervals differ?

Prior $\theta \sim \mathcal{N}(0,1)$. Confidence $C(b) = 2 \pm 1.96/\sqrt{3}$; credible $I(b) = 1.5 \pm 1.96(.5)$. Why different?

A. Narrower credible interval means better coverage.

B. The zero-centered prior adds information.

C. The physical sensor became less noisy.

Confidence interval; credible interval — [3] NIST §7.1.4 · [4] MIT 18.05 §2 · [1] DL §5.6.1 · Course comparison

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Transfer

Calibration: new data, new prior

Readings $b = (2, 3)^T$ estimate offset θ; iid noise $\mathcal{N}(0, 1)$; prior $\theta \sim \mathcal{N}(4, 1)$. Which is valid?

A. The prior changes the data-only MLE.

B. The prior center can be ignored in MAP.

C. MAP moves above the data mean; MLE stays fixed.

MLE; MAP; confidence and credible intervals — [1] DL §§5.5–5.6 · [3] NIST §7.1.4 · [4] MIT 18.05 §2 · Course question

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Intervals

What does the sensor promise?

A specification says “±2 cm, 95%.” Which conclusion is justified without another definition?

A. It is a confidence interval for the offset.

B. It is a credible interval for the offset.

C. None: ask what quantity and probability model it describes.

Confidence; credibility; predictive uncertainty — [3] NIST §7.1.4 · [4] MIT 18.05 §2 · [1] DL §5.6, Eq.5.68 · Course question

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Synthesis

Which modeling bridge is valid?

Zero-mean iid Gaussian noise; isotropic Gaussian prior. Which chain gives the negative-log posterior?

- **A** Independence gives squares; Gaussian noise gives sums.
- **B** Gaussian noise gives squares; independence gives sums.
- **C** A Gaussian prior removes the data residual term.

Gaussian likelihood; independence; MAP regularization — [1] DL §§5.5–5.6 · [1] DL §3.7 · Course question

References

Sources and optional extensions

[1] Goodfellow et al. Deep Learning. MIT Press, 2016.

[2] UBC CS340. Multivariate Gaussian Distributions and LDA. 2011.

[3] NIST/SEMATECH. e-Handbook of Statistical Methods.

[4] Orloff & Bloom. Probability Intervals. MIT 18.05, 2022.

[5] Welch & Bishop. An Introduction to the Kalman Filter. 2004.